

In this appendix we provide additional details on the designs of Section 6. Across all designs that appear in Section 6, Z is chosen to be uniformly distributed on $\mathcal{Z} = \{0, 1, 2\}$. Z is uniformly distributed on $\mathcal{Z} = \{0, 1\}$ for Table 9 in Appendix E.

Table S.1 presents the $Q \in \mathbf{Q}_{1s}$ used in the simulation of Section 6.1. For each row in Table S.1, the first four columns $Q(a_0, a_1, a_2, b_0, b_1, b_2)$ denote $Q\{Y(0) = a_0, Y(1) = a_1, Y(2) = a_2, D(0) = b_0, D(1) = b_1, D(2) = b_2\}$ for $(a_0, a_1, a_2, b_0, b_1, b_2) \in \{-1, 0, 1\}^3 \times \{0, 1, 2\}^3$, and the last four columns represent the probabilities assigned to the four events. The sum of the last four columns equals one.

The distribution in Table S.1 is also employed to construct the $Q \in \mathbf{Q}_{1s}$ with continuous outcomes discussed in Section 6.2. In particular, Q is generated as a mixture of truncated normal distributions where the mixture weights are given by a distribution $\tilde{Q}$ following Table S.1. Specifically, we let $\Phi_{\mathcal{Y}}(\cdot)$ denote the truncation of the normal distribution $N(\cdot, 1)$ on $\mathcal{Y}$. Each row in Table S.1 now becomes the probability $\tilde{Q}\{\mu(0) = a_0, \mu(1) = a_1, \mu(2) = a_2, D(0) = b_0, D(1) = b_1, D(2) = b_2\}$ of each mixture component. For each mixture component, $D(z) \equiv b_z$ for $z \in \mathcal{Z}$, and independently across $d \in \mathcal{D}$, $Y(d)$ is drawn from the truncated normal distribution $\Phi_{\mathcal{Y}}(a_d)$.

Table S.2 specifies the encouragement design distribution $Q \in \mathbf{Q}_{enc}$ used in Section 6.1. For each row, the first five columns $Q(a_0, a_1, a_2, b_0, b_1, b_2)$ denote $Q\{Y(0) = a_0, Y(1) = a_1, Y(2) = a_2, D(0) = b_0, D(1) = b_1, D(2) = b_2\}$, for $(a_0, a_1, a_2, b_0, b_1, b_2) \in \{-1, 0, 1\}^3 \times \{0, 1, 2\}^3$, and the last five columns represent the probabilities assigned to the five events. The sum of the last five columns equals one.

Table S.3 specifies the encouragement design distribution $Q \in \mathbf{Q}_{enc}$ used in Section 6.3 for $\mathcal{Y} = \{-1, 0, 1, 2\}$. For each row, the first four columns $Q(a_0, a_1, a_2, b_0, b_1, b_2)$ denote $Q\{Y(0) = a_0, Y(1) = a_1, Y(2) = a_2, D(0) = b_0, D(1) = b_1, D(2) = b_2\}$, for $(a_0, a_1, a_2, a_3, b_0, b_1, b_2) \in \{-1, 0, 1, 2\}^3 \times \{0, 1, 2\}^3$, and the last four columns represent the probabilities assigned to the four events. The sum of the last four columns equals one.

Table S.4 presents the distribution Q used in the simulation in Table 9 in Appendix E when $\mathcal{Y} = \{0, 1, 2\}$. For each row, the first four columns $Q(y_0, y_1, d_0, d_1)$ denote any non-zero $Q\{Y(0) = y_0, Y(1) = y_1, D(0) = d_0, D(1) = d_1\}$ for $(y_0, y_1, d_0, d_1) \in \{0, 1, 2\}^2 \times \{0, 1\}^2$, and the last four columns represent the probabilities assigned to the four events. The sum of the last four columns equals one.

Table S.5 presents the distribution Q used in the simulation in Table 9 in Appendix E when $\mathcal{Y} = \{0, 1, 2, 3\}$. For each row, the first six columns $Q(y_0, y_1, d_0, d_1)$ denote any non-zero $Q\{Y(0) = y_0, Y(1) = y_1, D(0) = d_0, D(1) = d_1\}$ for $(y_0, y_1, d_0, d_1) \in \{0, 1, 2, 3\}^2 \times \{0, 1\}^2$, and the last six columns represent the probabilities assigned to the six events. The sum of the last six columns equals one.

Events $(a_0, a_1, a_2, b_0, b_1, b_2)$			Probabilities assigned					
$Q(-1, -1, -1, 0, 0, 0)$	$Q(-1, -1, -1, 0, 1, 0)$	$Q(-1, -1, -1, 0, 0, 2)$	$Q(-1, -1, -1, 0, 1, 2)$	0.0001	0.0001	0.0001	0.0001	0.0001
$Q(-1, -1, 0, 0, 0, 0)$	$Q(-1, -1, 0, 0, 1, 0)$	$Q(-1, -1, 0, 0, 0, 2)$	$Q(-1, -1, 0, 0, 1, 2)$	0.0001	0.0044	0.0067	0.0051	
$Q(-1, -1, 1, 0, 0, 0)$	$Q(-1, -1, 1, 0, 1, 0)$	$Q(-1, -1, 1, 0, 0, 2)$	$Q(-1, -1, 1, 0, 1, 2)$	0.0001	0.0063	0.0001	0.0183	
$Q(-1, 0, -1, 0, 0, 0)$	$Q(-1, 0, -1, 0, 1, 0)$	$Q(-1, 0, -1, 0, 0, 2)$	$Q(-1, 0, -1, 0, 1, 2)$	0.0001	0.0001	0.0009	0.0001	
$Q(-1, 0, 0, 0, 0, 0)$	$Q(-1, 0, 0, 0, 1, 0)$	$Q(-1, 0, 0, 0, 0, 2)$	$Q(-1, 0, 0, 0, 1, 2)$	0.0001	0.0001	0.0070	0.0001	
$Q(-1, 0, 1, 0, 0, 0)$	$Q(-1, 0, 1, 0, 1, 0)$	$Q(-1, 0, 1, 0, 0, 2)$	$Q(-1, 0, 1, 0, 1, 2)$	0.0001	0.0001	0.0105	0.0056	
$Q(-1, 1, -1, 0, 0, 0)$	$Q(-1, 1, -1, 0, 1, 0)$	$Q(-1, 1, -1, 0, 0, 2)$	$Q(-1, 1, -1, 0, 1, 2)$	0.0001	0.0118	0.0001	0.0820	
$Q(-1, 1, 0, 0, 0, 0)$	$Q(-1, 1, 0, 0, 1, 0)$	$Q(-1, 1, 0, 0, 0, 2)$	$Q(-1, 1, 0, 0, 1, 2)$	0.0001	0.0001	0.0001	0.0064	
$Q(-1, 1, 1, 0, 0, 0)$	$Q(-1, 1, 1, 0, 1, 0)$	$Q(-1, 1, 1, 0, 0, 2)$	$Q(-1, 1, 1, 0, 1, 2)$	0.0001	0.0022	0.0001	0.0001	
$Q(0, -1, -1, 0, 0, 0)$	$Q(0, -1, -1, 0, 1, 0)$	$Q(0, -1, -1, 0, 0, 2)$	$Q(0, -1, -1, 0, 1, 2)$	0.0045	0.0001	0.0012	0.0033	
$Q(0, -1, 0, 0, 0, 0)$	$Q(0, -1, 0, 0, 1, 0)$	$Q(0, -1, 0, 0, 0, 2)$	$Q(0, -1, 0, 0, 1, 2)$	0.0077	0.0304	0.0021	0.0280	
$Q(0, -1, 1, 0, 0, 0)$	$Q(0, -1, 1, 0, 1, 0)$	$Q(0, -1, 1, 0, 0, 2)$	$Q(0, -1, 1, 0, 1, 2)$	0.0001	0.0001	0.0004	0.0103	
$Q(0, 0, -1, 0, 0, 0)$	$Q(0, 0, -1, 0, 1, 0)$	$Q(0, 0, -1, 0, 0, 2)$	$Q(0, 0, -1, 0, 1, 2)$	0.0001	0.0001	0.0001	0.0001	
$Q(0, 0, 0, 0, 0, 0)$	$Q(0, 0, 0, 0, 1, 0)$	$Q(0, 0, 0, 0, 0, 2)$	$Q(0, 0, 0, 0, 1, 2)$	0.0001	0.0001	0.0001	0.0001	
$Q(0, 0, 1, 0, 0, 0)$	$Q(0, 0, 1, 0, 1, 0)$	$Q(0, 0, 1, 0, 0, 2)$	$Q(0, 0, 1, 0, 1, 2)$	0.0001	0.0067	0.0001	0.4248	
$Q(0, 1, -1, 0, 0, 0)$	$Q(0, 1, -1, 0, 1, 0)$	$Q(0, 1, -1, 0, 0, 2)$	$Q(0, 1, -1, 0, 1, 2)$	0.0001	0.0001	0.0001	0.0001	
$Q(0, 1, 0, 0, 0, 0)$	$Q(0, 1, 0, 0, 1, 0)$	$Q(0, 1, 0, 0, 0, 2)$	$Q(0, 1, 0, 0, 1, 2)$	0.0159	0.0001	0.0001	0.0001	
$Q(0, 1, 1, 0, 0, 0)$	$Q(0, 1, 1, 0, 1, 0)$	$Q(0, 1, 1, 0, 0, 2)$	$Q(0, 1, 1, 0, 1, 2)$	0.0001	0.0098	0.0001	0.0120	
$Q(1, -1, -1, 0, 0, 0)$	$Q(1, -1, -1, 0, 1, 0)$	$Q(1, -1, -1, 0, 0, 2)$	$Q(1, -1, -1, 0, 1, 2)$	0.0001	0.0360	0.0001	0.0677	
$Q(1, -1, 0, 0, 0, 0)$	$Q(1, -1, 0, 0, 1, 0)$	$Q(1, -1, 0, 0, 0, 2)$	$Q(1, -1, 0, 0, 1, 2)$	0.0008	0.0028	0.0003	0.0055	
$Q(1, -1, 1, 0, 0, 0)$	$Q(1, -1, 1, 0, 1, 0)$	$Q(1, -1, 1, 0, 0, 2)$	$Q(1, -1, 1, 0, 1, 2)$	0.0033	0.0001	0.0033	0.0001	
$Q(1, 0, -1, 0, 0, 0)$	$Q(1, 0, -1, 0, 1, 0)$	$Q(1, 0, -1, 0, 0, 2)$	$Q(1, 0, -1, 0, 1, 2)$	0.0056	0.0149	0.0001	0.0001	
$Q(1, 0, 0, 0, 0, 0)$	$Q(1, 0, 0, 0, 1, 0)$	$Q(1, 0, 0, 0, 0, 2)$	$Q(1, 0, 0, 0, 1, 2)$	0.0001	0.0403	0.0083	0.0029	
$Q(1, 0, 1, 0, 0, 0)$	$Q(1, 0, 1, 0, 1, 0)$	$Q(1, 0, 1, 0, 0, 2)$	$Q(1, 0, 1, 0, 1, 2)$	0.0008	0.0001	0.0001	0.0113	
$Q(1, 1, -1, 0, 0, 0)$	$Q(1, 1, -1, 0, 1, 0)$	$Q(1, 1, -1, 0, 0, 2)$	$Q(1, 1, -1, 0, 1, 2)$	0.0001	0.0001	0.0001	0.0056	
$Q(1, 1, 0, 0, 0, 0)$	$Q(1, 1, 0, 0, 1, 0)$	$Q(1, 1, 0, 0, 0, 2)$	$Q(1, 1, 0, 0, 1, 2)$	0.0001	0.0001	0.0227	0.0001	
$Q(1, 1, 1, 0, 0, 0)$	$Q(1, 1, 1, 0, 1, 0)$	$Q(1, 1, 1, 0, 0, 2)$	$Q(1, 1, 1, 0, 1, 2)$	0.0049	0.0001	0.0001	0.0325	

Table S.1: Distribution of $Q \in \mathbf{Q}_{1s}$.

Table S.3: Distribution of Q .

Events ($a_0, a_1, a_2, b_0, b_1, b_2$)				Probabilities assigned			
$Q(-1, -1, -1, 0, 0, 0)$	$Q(-1, -1, -1, 0, 0, 2)$	$Q(-1, -1, -1, 0, 1, 0)$	$Q(-1, -1, -1, 0, 1, 1)$	0.0015	0.0019	0.0017	0.0017
$Q(-1, -1, -1, 0, 1, 2)$	$Q(-1, -1, -1, 0, 2, 2)$	$Q(-1, -1, -1, 1, 1, 1)$	$Q(-1, -1, -1, 1, 1, 2)$	0.0017	0.0003	0.0014	0.0012
$Q(-1, -1, -1, 2, 1, 2)$	$Q(-1, -1, -1, 2, 2, 2)$	$Q(-1, -1, 0, 0, 0, 0)$	$Q(-1, -1, 0, 0, 0, 2)$	0.0017	0.0016	0.0015	0.0018
$Q(-1, -1, 0, 0, 1, 0)$	$Q(-1, -1, 0, 0, 1, 1)$	$Q(-1, -1, 0, 0, 1, 2)$	$Q(-1, -1, 0, 0, 2, 2)$	0.0017	0.0012	0.0014	0.0021
$Q(-1, -1, 0, 1, 1, 1)$	$Q(-1, -1, 0, 1, 1, 2)$	$Q(-1, -1, 0, 2, 1, 2)$	$Q(-1, -1, 0, 2, 2, 2)$	0.0018	0.0015	0.0009	0.0012
$Q(-1, -1, 1, 0, 0, 0)$	$Q(-1, -1, 1, 0, 0, 2)$	$Q(-1, -1, 1, 0, 1, 0)$	$Q(-1, -1, 1, 0, 1, 1)$	0.0019	0.0020	0.0014	0.0018
$Q(-1, -1, 1, 0, 1, 2)$	$Q(-1, -1, 1, 0, 2, 2)$	$Q(-1, -1, 1, 1, 1, 1)$	$Q(-1, -1, 1, 1, 1, 2)$	0.0004	0.0018	0.0016	0.0013
$Q(-1, -1, 1, 2, 1, 2)$	$Q(-1, -1, 1, 2, 2, 2)$	$Q(-1, -1, 2, 0, 0, 0)$	$Q(-1, -1, 2, 0, 0, 2)$	0.0018	0.0008	0.0016	0.0018
$Q(-1, -1, 2, 0, 1, 0)$	$Q(-1, -1, 2, 0, 1, 1)$	$Q(-1, -1, 2, 0, 1, 2)$	$Q(-1, -1, 2, 0, 2, 2)$	0.0020	0.0014	0.0013	0.0016
$Q(-1, -1, 2, 1, 1, 1)$	$Q(-1, -1, 2, 1, 1, 2)$	$Q(-1, -1, 2, 2, 1, 2)$	$Q(-1, -1, 2, 2, 2, 2)$	0.0016	0.0020	0.0016	0.0020
$Q(-1, 0, -1, 0, 0, 0)$	$Q(-1, 0, -1, 0, 0, 2)$	$Q(-1, 0, -1, 0, 1, 0)$	$Q(-1, 0, -1, 0, 1, 1)$	0.0017	0.0015	0.0016	0.0014
$Q(-1, 0, -1, 0, 1, 2)$	$Q(-1, 0, -1, 0, 2, 2)$	$Q(-1, 0, -1, 1, 1, 1)$	$Q(-1, 0, -1, 1, 1, 2)$	0.0018	0.0018	0.0004	0.0020
$Q(-1, 0, -1, 2, 1, 2)$	$Q(-1, 0, -1, 2, 2, 2)$	$Q(-1, 0, 0, 0, 0, 0)$	$Q(-1, 0, 0, 0, 0, 2)$	0.0015	0.0016	0.0011	0.0019
$Q(-1, 0, 0, 0, 1, 0)$	$Q(-1, 0, 0, 0, 1, 1)$	$Q(-1, 0, 0, 0, 1, 2)$	$Q(-1, 0, 0, 0, 2, 2)$	0.0017	0.0022	0.0016	0.0019
$Q(-1, 0, 0, 1, 1, 1)$	$Q(-1, 0, 0, 1, 1, 2)$	$Q(-1, 0, 0, 2, 1, 2)$	$Q(-1, 0, 0, 2, 2, 2)$	0.0019	0.0019	0.0014	0.0018
$Q(-1, 0, 1, 0, 0, 0)$	$Q(-1, 0, 1, 0, 0, 2)$	$Q(-1, 0, 1, 0, 1, 0)$	$Q(-1, 0, 1, 0, 1, 1)$	0.0017	0.0021	0.0022	0.0018
$Q(-1, 0, 1, 0, 1, 2)$	$Q(-1, 0, 1, 0, 2, 2)$	$Q(-1, 0, 1, 1, 1, 1)$	$Q(-1, 0, 1, 1, 1, 2)$	0.0019	0.0014	0.0019	0.0010
$Q(-1, 0, 1, 2, 1, 2)$	$Q(-1, 0, 1, 2, 2, 2)$	$Q(-1, 0, 2, 0, 0, 0)$	$Q(-1, 0, 2, 0, 0, 2)$	0.0019	0.0009	0.0021	0.0007
$Q(-1, 0, 2, 0, 1, 0)$	$Q(-1, 0, 2, 0, 1, 1)$	$Q(-1, 0, 2, 0, 1, 2)$	$Q(-1, 0, 2, 0, 2, 2)$	0.0016	0.0017	0.0008	0.0015
$Q(-1, 0, 2, 1, 1, 1)$	$Q(-1, 0, 2, 1, 1, 2)$	$Q(-1, 0, 2, 2, 1, 2)$	$Q(-1, 0, 2, 2, 2, 2)$	0.0013	0.0014	0.0014	0.0011
$Q(-1, 1, -1, 0, 0, 0)$	$Q(-1, 1, -1, 0, 0, 2)$	$Q(-1, 1, -1, 0, 1, 0)$	$Q(-1, 1, -1, 0, 1, 1)$	0.0012	0.0020	0.0017	0.0014
$Q(-1, 1, -1, 0, 1, 2)$	$Q(-1, 1, -1, 0, 2, 2)$	$Q(-1, 1, -1, 1, 1, 1)$	$Q(-1, 1, -1, 1, 1, 2)$	0.0021	0.0018	0.0012	0.0014
$Q(-1, 1, -1, 2, 1, 2)$	$Q(-1, 1, -1, 2, 2, 2)$	$Q(-1, 1, 0, 0, 0, 0)$	$Q(-1, 1, 0, 0, 0, 2)$	0.0015	0.0012	0.0012	0.0016
$Q(-1, 1, 0, 0, 1, 0)$	$Q(-1, 1, 0, 0, 1, 1)$	$Q(-1, 1, 0, 0, 1, 2)$	$Q(-1, 1, 0, 0, 2, 2)$	0.0015	0.0011	0.0019	0.0011
$Q(-1, 1, 0, 1, 1, 1)$	$Q(-1, 1, 0, 1, 1, 2)$	$Q(-1, 1, 0, 2, 1, 2)$	$Q(-1, 1, 0, 2, 2, 2)$	0.0021	0.0019	0.0014	0.0017
$Q(-1, 1, 1, 0, 0, 0)$	$Q(-1, 1, 1, 0, 0, 2)$	$Q(-1, 1, 1, 0, 1, 0)$	$Q(-1, 1, 1, 0, 1, 1)$	0.0017	0.0012	0.0010	0.0019
$Q(-1, 1, 1, 0, 1, 2)$	$Q(-1, 1, 1, 0, 2, 2)$	$Q(-1, 1, 1, 1, 1, 1)$	$Q(-1, 1, 1, 1, 1, 2)$	0.0018	0.0021	0.0017	0.0009
$Q(-1, 1, 1, 2, 1, 2)$	$Q(-1, 1, 1, 2, 2, 2)$	$Q(-1, 1, 2, 0, 0, 0)$	$Q(-1, 1, 2, 0, 0, 2)$	0.0016	0.0019	0.0012	0.0015
$Q(-1, 1, 2, 0, 1, 0)$	$Q(-1, 1, 2, 0, 1, 1)$	$Q(-1, 1, 2, 0, 1, 2)$	$Q(-1, 1, 2, 0, 2, 2)$	0.0013	0.0014	0.0012	0.0016
$Q(-1, 1, 2, 1, 1, 1)$	$Q(-1, 1, 2, 1, 1, 2)$	$Q(-1, 1, 2, 2, 1, 2)$	$Q(-1, 1, 2, 2, 2, 2)$	0.0016	0.0018	0.0010	0.0014
$Q(-1, 2, -1, 0, 0, 0)$	$Q(-1, 2, -1, 0, 0, 2)$	$Q(-1, 2, -1, 0, 1, 0)$	$Q(-1, 2, -1, 0, 1, 1)$	0.0013	0.0021	0.0017	0.0014
$Q(-1, 2, -1, 0, 1, 2)$	$Q(-1, 2, -1, 0, 2, 2)$	$Q(-1, 2, -1, 1, 1, 1)$	$Q(-1, 2, -1, 1, 1, 2)$	0.0010	0.0019	0.0022	0.0018
$Q(-1, 2, -1, 2, 1, 2)$	$Q(-1, 2, -1, 2, 2, 2)$	$Q(-1, 2, 0, 0, 0, 0)$	$Q(-1, 2, 0, 0, 0, 2)$	0.0013	0.0017	0.0021	0.0019

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Table S.3: Distribution of Q (continued).

Events ($a_0, a_1, a_2, b_0, b_1, b_2$)			Probabilities assigned									
$Q(-1, 2, 0, 0, 1, 0)$	$Q(-1, 2, 0, 0, 1, 1)$	$Q(-1, 2, 0, 0, 1, 2)$	$Q(-1, 2, 0, 0, 2, 2)$	0.0010	0.0007	0.0017	0.0013					
$Q(-1, 2, 0, 1, 1, 1)$	$Q(-1, 2, 0, 1, 1, 2)$	$Q(-1, 2, 0, 2, 1, 2)$	$Q(-1, 2, 0, 2, 2, 2)$	0.0012	0.0008	0.0022	0.0016					
$Q(-1, 2, 1, 0, 0, 0)$	$Q(-1, 2, 1, 0, 0, 2)$	$Q(-1, 2, 1, 0, 1, 0)$	$Q(-1, 2, 1, 0, 1, 1)$	0.0007	0.0015	0.0014	0.0010					
$Q(-1, 2, 1, 0, 1, 2)$	$Q(-1, 2, 1, 0, 2, 2)$	$Q(-1, 2, 1, 1, 1, 1)$	$Q(-1, 2, 1, 1, 1, 2)$	0.0014	0.0016	0.0020	0.0019					
$Q(-1, 2, 1, 2, 1, 2)$	$Q(-1, 2, 1, 2, 2, 2)$	$Q(-1, 2, 2, 0, 0, 0)$	$Q(-1, 2, 2, 0, 0, 2)$	0.0015	0.0010	0.0021	0.0010					
$Q(-1, 2, 2, 0, 1, 0)$	$Q(-1, 2, 2, 0, 1, 1)$	$Q(-1, 2, 2, 0, 1, 2)$	$Q(-1, 2, 2, 0, 2, 2)$	0.0006	0.0011	0.0020	0.0018					
$Q(-1, 2, 2, 1, 1, 1)$	$Q(-1, 2, 2, 1, 1, 2)$	$Q(-1, 2, 2, 2, 1, 2)$	$Q(-1, 2, 2, 2, 2, 2)$	0.0019	0.0015	0.0019	0.0022					
$Q(0, -1, -1, 0, 0, 0)$	$Q(0, -1, -1, 0, 0, 2)$	$Q(0, -1, -1, 0, 1, 0)$	$Q(0, -1, -1, 0, 1, 1)$	0.0016	0.0014	0.0015	0.0007					
$Q(0, -1, -1, 0, 1, 2)$	$Q(0, -1, -1, 0, 2, 2)$	$Q(0, -1, -1, 1, 1, 1, 1)$	$Q(0, -1, -1, 1, 1, 2)$	0.0010	0.0012	0.0016	0.0020					
$Q(0, -1, -1, 2, 1, 2)$	$Q(0, -1, -1, 2, 2, 2)$	$Q(0, -1, 0, 0, 0, 0)$	$Q(0, -1, 0, 0, 0, 2)$	0.0021	0.0015	0.0008	0.0013					
$Q(0, -1, 0, 0, 1, 0)$	$Q(0, -1, 0, 0, 1, 1)$	$Q(0, -1, 0, 0, 1, 2)$	$Q(0, -1, 0, 0, 2, 2)$	0.0017	0.0015	0.0015	0.0019					
$Q(0, -1, 0, 1, 1, 1)$	$Q(0, -1, 0, 1, 1, 2)$	$Q(0, -1, 0, 2, 1, 2)$	$Q(0, -1, 0, 2, 2, 2)$	0.0013	0.0008	0.0014	0.0017					
$Q(0, -1, 1, 0, 0, 0)$	$Q(0, -1, 1, 0, 0, 2)$	$Q(0, -1, 1, 0, 1, 0)$	$Q(0, -1, 1, 0, 1, 1)$	0.0007	0.0005	0.0014	0.0014					
$Q(0, -1, 1, 0, 1, 2)$	$Q(0, -1, 1, 0, 2, 2)$	$Q(0, -1, 1, 1, 1, 1)$	$Q(0, -1, 1, 1, 1, 2)$	0.0012	0.0023	0.0013	0.0022					
$Q(0, -1, 1, 2, 1, 2)$	$Q(0, -1, 1, 2, 2, 2)$	$Q(0, -1, 2, 0, 0, 0)$	$Q(0, -1, 2, 0, 0, 2)$	0.0011	0.0019	0.0021	0.0017					
$Q(0, -1, 2, 0, 1, 0)$	$Q(0, -1, 2, 0, 1, 1)$	$Q(0, -1, 2, 0, 1, 2)$	$Q(0, -1, 2, 0, 2, 2)$	0.0007	0.0020	0.0018	0.0022					
$Q(0, -1, 2, 1, 1, 1)$	$Q(0, -1, 2, 1, 1, 2)$	$Q(0, -1, 2, 2, 1, 2)$	$Q(0, -1, 2, 2, 2, 2)$	0.0011	0.0018	0.0019	0.0020					
$Q(0, 0, -1, 0, 0, 0)$	$Q(0, 0, -1, 0, 0, 2)$	$Q(0, 0, -1, 0, 1, 0)$	$Q(0, 0, -1, 0, 1, 1)$	0.0013	0.0022	0.0019	0.0015					
$Q(0, 0, -1, 0, 1, 2)$	$Q(0, 0, -1, 0, 2, 2)$	$Q(0, 0, -1, 1, 1, 1)$	$Q(0, 0, -1, 1, 1, 2)$	0.0018	0.0011	0.0016	0.0016					
$Q(0, 0, -1, 2, 1, 2)$	$Q(0, 0, -1, 2, 2, 2)$	$Q(0, 0, 0, 0, 0, 0)$	$Q(0, 0, 0, 0, 0, 2)$	0.0017	0.0021	0.0007	0.0009					
$Q(0, 0, 0, 0, 1, 0)$	$Q(0, 0, 0, 0, 1, 1)$	$Q(0, 0, 0, 0, 1, 2)$	$Q(0, 0, 0, 0, 2, 2)$	0.0014	0.0016	0.0021	0.0019					
$Q(0, 0, 0, 1, 1, 1)$	$Q(0, 0, 0, 1, 1, 2)$	$Q(0, 0, 0, 2, 1, 2)$	$Q(0, 0, 0, 2, 2, 2)$	0.0013	0.0017	0.0020	0.0011					
$Q(0, 0, 1, 0, 0, 0)$	$Q(0, 0, 1, 0, 0, 2)$	$Q(0, 0, 1, 0, 1, 0)$	$Q(0, 0, 1, 0, 1, 1)$	0.0006	0.0013	0.0018	0.0016					
$Q(0, 0, 1, 0, 1, 2)$	$Q(0, 0, 1, 0, 2, 2)$	$Q(0, 0, 1, 1, 1, 1)$	$Q(0, 0, 1, 1, 1, 2)$	0.0014	0.0017	0.0019	0.0021					
$Q(0, 0, 1, 2, 1, 2)$	$Q(0, 0, 1, 2, 2, 2)$	$Q(0, 0, 2, 0, 0, 0)$	$Q(0, 0, 2, 0, 0, 2)$	0.0012	0.0018	0.0018	0.0021					
$Q(0, 0, 2, 0, 1, 0)$	$Q(0, 0, 2, 0, 1, 1)$	$Q(0, 0, 2, 0, 1, 2)$	$Q(0, 0, 2, 0, 2, 2)$	0.0019	0.0005	0.0018	0.0018					
$Q(0, 0, 2, 1, 1, 1)$	$Q(0, 0, 2, 1, 1, 2)$	$Q(0, 0, 2, 2, 1, 2)$	$Q(0, 0, 2, 2, 2, 2)$	0.0017	0.0013	0.0006	0.0016					
$Q(0, 1, -1, 0, 0, 0)$	$Q(0, 1, -1, 0, 0, 2)$	$Q(0, 1, -1, 0, 1, 0)$	$Q(0, 1, -1, 0, 1, 1)$	0.0017	0.0021	0.0017	0.0019					
$Q(0, 1, -1, 0, 1, 2)$	$Q(0, 1, -1, 0, 2, 2)$	$Q(0, 1, -1, 1, 1, 1, 1)$	$Q(0, 1, -1, 1, 1, 2)$	0.0012	0.0010	0.0020	0.0016					
$Q(0, 1, -1, 2, 1, 2)$	$Q(0, 1, -1, 2, 2, 2)$	$Q(0, 1, 0, 0, 0, 0)$	$Q(0, 1, 0, 0, 0, 2)$	0.0019	0.0020	0.0011	0.0006					
$Q(0, 1, 0, 0, 1, 0)$	$Q(0, 1, 0, 0, 1, 1)$	$Q(0, 1, 0, 0, 1, 2)$	$Q(0, 1, 0, 0, 2, 2)$	0.0017	0.0010	0.0021	0.0014					
$Q(0, 1, 0, 1, 1, 1)$	$Q(0, 1, 0, 1, 1, 2)$	$Q(0, 1, 0, 2, 1, 2)$	$Q(0, 1, 0, 2, 2, 2)$	0.0015	0.0019	0.0013	0.0020					
$Q(0, 1, 1, 0, 0, 0)$	$Q(0, 1, 1, 0, 0, 2)$	$Q(0, 1, 1, 0, 1, 0)$	$Q(0, 1, 1, 0, 1, 1)$	0.0010	0.0023	0.0019	0.0021					

Continued on next page

Table S.3: Distribution of Q (continued).

Events ($a_0, a_1, a_2, b_0, b_1, b_2$)		Probabilities assigned	
$Q(0, 1, 1, 0, 1, 2)$	$Q(0, 1, 1, 0, 2, 2)$	$Q(0, 1, 1, 1, 1, 1)$	$Q(0, 1, 1, 1, 1, 2)$
$Q(0, 1, 1, 2, 1, 2)$	$Q(0, 1, 1, 2, 2, 2)$	$Q(0, 1, 2, 0, 0, 0)$	$Q(0, 1, 2, 0, 0, 2)$
$Q(0, 1, 2, 0, 1, 0)$	$Q(0, 1, 2, 0, 1, 1)$	$Q(0, 1, 2, 0, 1, 2)$	$Q(0, 1, 2, 0, 2, 2)$
$Q(0, 1, 2, 1, 1, 1)$	$Q(0, 1, 2, 1, 1, 2)$	$Q(0, 1, 2, 2, 1, 2)$	$Q(0, 1, 2, 2, 2, 2)$
$Q(0, 2, -1, 0, 0, 0)$	$Q(0, 2, -1, 0, 0, 2)$	$Q(0, 2, -1, 0, 1, 0)$	$Q(0, 2, -1, 0, 1, 1)$
$Q(0, 2, -1, 0, 1, 2)$	$Q(0, 2, -1, 0, 2, 2)$	$Q(0, 2, -1, 1, 1, 1)$	$Q(0, 2, -1, 1, 1, 2)$
$Q(0, 2, -1, 2, 1, 2)$	$Q(0, 2, -1, 2, 2, 2)$	$Q(0, 2, 0, 0, 0, 0)$	$Q(0, 2, 0, 0, 0, 2)$
$Q(0, 2, 0, 0, 1, 0)$	$Q(0, 2, 0, 0, 1, 1)$	$Q(0, 2, 0, 0, 1, 2)$	$Q(0, 2, 0, 0, 2, 2)$
$Q(0, 2, 0, 1, 1, 1)$	$Q(0, 2, 0, 1, 1, 2)$	$Q(0, 2, 0, 2, 1, 2)$	$Q(0, 2, 0, 2, 2, 2)$
$Q(0, 2, 1, 0, 0, 0)$	$Q(0, 2, 1, 0, 0, 2)$	$Q(0, 2, 1, 0, 1, 0)$	$Q(0, 2, 1, 0, 1, 1)$
$Q(0, 2, 1, 0, 1, 2)$	$Q(0, 2, 1, 0, 2, 2)$	$Q(0, 2, 1, 1, 1, 1)$	$Q(0, 2, 1, 1, 1, 2)$
$Q(0, 2, 1, 2, 1, 2)$	$Q(0, 2, 1, 2, 2, 2)$	$Q(0, 2, 2, 0, 0, 0)$	$Q(0, 2, 2, 0, 0, 2)$
$Q(0, 2, 2, 0, 1, 0)$	$Q(0, 2, 2, 0, 1, 1)$	$Q(0, 2, 2, 0, 1, 2)$	$Q(0, 2, 2, 0, 2, 2)$
$Q(0, 2, 2, 1, 1, 1)$	$Q(0, 2, 2, 1, 1, 2)$	$Q(0, 2, 2, 2, 1, 2)$	$Q(0, 2, 2, 2, 2, 2)$
$Q(1, -1, -1, 0, 0, 0)$	$Q(1, -1, -1, 0, 0, 2)$	$Q(1, -1, -1, 0, 1, 0)$	$Q(1, -1, -1, 0, 1, 1)$
$Q(1, -1, -1, 0, 1, 2)$	$Q(1, -1, -1, 0, 2, 2)$	$Q(1, -1, -1, 1, 1, 1)$	$Q(1, -1, -1, 1, 1, 2)$
$Q(1, -1, -1, 2, 1, 2)$	$Q(1, -1, -1, 2, 2, 2)$	$Q(1, -1, 0, 0, 0, 0)$	$Q(1, -1, 0, 0, 0, 2)$
$Q(1, -1, 0, 0, 1, 0)$	$Q(1, -1, 0, 0, 1, 1)$	$Q(1, -1, 0, 0, 1, 2)$	$Q(1, -1, 0, 0, 2, 2)$
$Q(1, -1, 0, 1, 1, 1)$	$Q(1, -1, 0, 1, 1, 2)$	$Q(1, -1, 0, 2, 1, 2)$	$Q(1, -1, 0, 2, 2, 2)$
$Q(1, -1, 1, 0, 0, 0)$	$Q(1, -1, 1, 0, 0, 2)$	$Q(1, -1, 1, 0, 1, 0)$	$Q(1, -1, 1, 0, 1, 1)$
$Q(1, -1, 1, 0, 1, 2)$	$Q(1, -1, 1, 0, 2, 2)$	$Q(1, -1, 1, 1, 1, 1)$	$Q(1, -1, 1, 1, 1, 2)$
$Q(1, -1, 1, 2, 1, 2)$	$Q(1, -1, 1, 2, 2, 2)$	$Q(1, -1, 2, 0, 0, 0)$	$Q(1, -1, 2, 0, 0, 2)$
$Q(1, -1, 2, 0, 1, 0)$	$Q(1, -1, 2, 0, 1, 1)$	$Q(1, -1, 2, 0, 1, 2)$	$Q(1, -1, 2, 0, 2, 2)$
$Q(1, -1, 2, 1, 1, 1)$	$Q(1, -1, 2, 1, 1, 2)$	$Q(1, -1, 2, 2, 1, 2)$	$Q(1, -1, 2, 2, 2, 2)$
$Q(1, 0, -1, 0, 0, 0)$	$Q(1, 0, -1, 0, 0, 2)$	$Q(1, 0, -1, 0, 1, 0)$	$Q(1, 0, -1, 0, 1, 1)$
$Q(1, 0, -1, 0, 1, 2)$	$Q(1, 0, -1, 0, 2, 2)$	$Q(1, 0, -1, 1, 1, 1)$	$Q(1, 0, -1, 1, 1, 2)$
$Q(1, 0, -1, 2, 1, 2)$	$Q(1, 0, -1, 2, 2, 2)$	$Q(1, 0, 0, 0, 0, 0)$	$Q(1, 0, 0, 0, 0, 2)$
$Q(1, 0, 0, 0, 1, 0)$	$Q(1, 0, 0, 0, 1, 1)$	$Q(1, 0, 0, 0, 1, 2)$	$Q(1, 0, 0, 0, 2, 2)$
$Q(1, 0, 0, 1, 1, 1)$	$Q(1, 0, 0, 1, 1, 2)$	$Q(1, 0, 0, 2, 1, 2)$	$Q(1, 0, 0, 2, 2, 2)$
$Q(1, 0, 1, 0, 0, 0)$	$Q(1, 0, 1, 0, 0, 2)$	$Q(1, 0, 1, 0, 1, 0)$	$Q(1, 0, 1, 0, 1, 1)$
$Q(1, 0, 1, 0, 1, 2)$	$Q(1, 0, 1, 0, 2, 2)$	$Q(1, 0, 1, 1, 1, 1)$	$Q(1, 0, 1, 1, 1, 2)$
$Q(1, 0, 1, 2, 1, 2)$	$Q(1, 0, 1, 2, 2, 2)$	$Q(1, 0, 2, 0, 0, 0)$	$Q(1, 0, 2, 0, 0, 2)$
$Q(1, 0, 2, 0, 1, 0)$	$Q(1, 0, 2, 0, 1, 1)$	$Q(1, 0, 2, 0, 1, 2)$	$Q(1, 0, 2, 0, 2, 2)$

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Table S.3: Distribution of Q (continued).

Events ($a_0, a_1, a_2, b_0, b_1, b_2$)			Probabilities assigned	
$Q(1, 0, 2, 1, 1, 1)$	$Q(1, 0, 2, 1, 1, 2)$	$Q(1, 0, 2, 2, 1, 2)$	$Q(1, 0, 2, 2, 2, 2)$	0.0011 0.0008 0.0016 0.0012
$Q(1, 1, -1, 0, 0, 0)$	$Q(1, 1, -1, 0, 0, 2)$	$Q(1, 1, -1, 0, 1, 0)$	$Q(1, 1, -1, 0, 1, 1)$	0.0022 0.0010 0.0010 0.0017
$Q(1, 1, -1, 0, 1, 2)$	$Q(1, 1, -1, 0, 2, 2)$	$Q(1, 1, -1, 1, 1, 1)$	$Q(1, 1, -1, 1, 1, 2)$	0.0014 0.0014 0.0013 0.0013
$Q(1, 1, -1, 2, 1, 2)$	$Q(1, 1, -1, 2, 2, 2)$	$Q(1, 1, 0, 0, 0, 0)$	$Q(1, 1, 0, 0, 0, 2)$	0.0019 0.0015 0.0021 0.0016
$Q(1, 1, 0, 0, 1, 0)$	$Q(1, 1, 0, 0, 1, 1)$	$Q(1, 1, 0, 0, 1, 2)$	$Q(1, 1, 0, 0, 2, 2)$	0.0016 0.0018 0.0012 0.0016
$Q(1, 1, 0, 1, 1, 1)$	$Q(1, 1, 0, 1, 1, 2)$	$Q(1, 1, 0, 2, 1, 2)$	$Q(1, 1, 0, 2, 2, 2)$	0.0021 0.0015 0.0013 0.0017
$Q(1, 1, 1, 0, 0, 0)$	$Q(1, 1, 1, 0, 0, 2)$	$Q(1, 1, 1, 0, 1, 0)$	$Q(1, 1, 1, 0, 1, 1)$	0.0016 0.0014 0.0017 0.0022
$Q(1, 1, 1, 0, 1, 2)$	$Q(1, 1, 1, 0, 2, 2)$	$Q(1, 1, 1, 1, 1, 1)$	$Q(1, 1, 1, 1, 1, 2)$	0.0015 0.0016 0.0015 0.0011
$Q(1, 1, 1, 2, 1, 2)$	$Q(1, 1, 1, 2, 2, 2)$	$Q(1, 1, 2, 0, 0, 0)$	$Q(1, 1, 2, 0, 0, 2)$	0.0017 0.0011 0.0015 0.0016
$Q(1, 1, 2, 0, 1, 0)$	$Q(1, 1, 2, 0, 1, 1)$	$Q(1, 1, 2, 0, 1, 2)$	$Q(1, 1, 2, 0, 2, 2)$	0.0015 0.0013 0.0014 0.0014
$Q(1, 1, 2, 1, 1, 1)$	$Q(1, 1, 2, 1, 1, 2)$	$Q(1, 1, 2, 2, 1, 2)$	$Q(1, 1, 2, 2, 2, 2)$	0.0016 0.0016 0.0009 0.0023
$Q(1, 2, -1, 0, 0, 0)$	$Q(1, 2, -1, 0, 0, 2)$	$Q(1, 2, -1, 0, 1, 0)$	$Q(1, 2, -1, 0, 1, 1)$	0.0014 0.0021 0.0017 0.0004
$Q(1, 2, -1, 0, 1, 2)$	$Q(1, 2, -1, 0, 2, 2)$	$Q(1, 2, -1, 1, 1, 1)$	$Q(1, 2, -1, 1, 1, 2)$	0.0011 0.0018 0.0016 0.0009
$Q(1, 2, -1, 2, 1, 2)$	$Q(1, 2, -1, 2, 2, 2)$	$Q(1, 2, 0, 0, 0, 0)$	$Q(1, 2, 0, 0, 0, 2)$	0.0013 0.0007 0.0016 0.0020
$Q(1, 2, 0, 0, 1, 0)$	$Q(1, 2, 0, 0, 1, 1)$	$Q(1, 2, 0, 0, 1, 2)$	$Q(1, 2, 0, 0, 2, 2)$	0.0020 0.0012 0.0020 0.0013
$Q(1, 2, 0, 1, 1, 1)$	$Q(1, 2, 0, 1, 1, 2)$	$Q(1, 2, 0, 2, 1, 2)$	$Q(1, 2, 0, 2, 2, 2)$	0.0016 0.0012 0.0021 0.0020
$Q(1, 2, 1, 0, 0, 0)$	$Q(1, 2, 1, 0, 0, 2)$	$Q(1, 2, 1, 0, 1, 0)$	$Q(1, 2, 1, 0, 1, 1)$	0.0013 0.0017 0.0012 0.0020
$Q(1, 2, 1, 0, 1, 2)$	$Q(1, 2, 1, 0, 2, 2)$	$Q(1, 2, 1, 1, 1, 1)$	$Q(1, 2, 1, 1, 1, 2)$	0.0015 0.0015 0.0020 0.0014
$Q(1, 2, 1, 2, 1, 2)$	$Q(1, 2, 1, 2, 2, 2)$	$Q(1, 2, 2, 0, 0, 0)$	$Q(1, 2, 2, 0, 0, 2)$	0.0013 0.0014 0.0010 0.0007
$Q(1, 2, 2, 0, 1, 0)$	$Q(1, 2, 2, 0, 1, 1)$	$Q(1, 2, 2, 0, 1, 2)$	$Q(1, 2, 2, 0, 2, 2)$	0.0018 0.0022 0.0019 0.0013
$Q(1, 2, 2, 1, 1, 1)$	$Q(1, 2, 2, 1, 1, 2)$	$Q(1, 2, 2, 2, 1, 2)$	$Q(1, 2, 2, 2, 2, 2)$	0.0015 0.0020 0.0012 0.0021
$Q(2, -1, -1, 0, 0, 0)$	$Q(2, -1, -1, 0, 0, 2)$	$Q(2, -1, -1, 0, 1, 0)$	$Q(2, -1, -1, 0, 1, 1)$	0.0018 0.0020 0.0010 0.0012
$Q(2, -1, -1, 0, 1, 2)$	$Q(2, -1, -1, 0, 2, 2)$	$Q(2, -1, -1, 1, 1, 1)$	$Q(2, -1, -1, 1, 1, 2)$	0.0015 0.0020 0.0017 0.0017
$Q(2, -1, -1, 2, 1, 2)$	$Q(2, -1, -1, 2, 2, 2)$	$Q(2, -1, 0, 0, 0, 0)$	$Q(2, -1, 0, 0, 0, 2)$	0.0011 0.0022 0.0015 0.0014
$Q(2, -1, 0, 0, 1, 0)$	$Q(2, -1, 0, 0, 1, 1)$	$Q(2, -1, 0, 0, 1, 2)$	$Q(2, -1, 0, 0, 2, 2)$	0.0021 0.0015 0.0011 0.0018
$Q(2, -1, 0, 1, 1, 1)$	$Q(2, -1, 0, 1, 1, 2)$	$Q(2, -1, 0, 2, 1, 2)$	$Q(2, -1, 0, 2, 2, 2)$	0.0015 0.0012 0.0021 0.0016
$Q(2, -1, 1, 0, 0, 0)$	$Q(2, -1, 1, 0, 0, 2)$	$Q(2, -1, 1, 0, 1, 0)$	$Q(2, -1, 1, 0, 1, 1)$	0.0020 0.0021 0.0017 0.0014
$Q(2, -1, 1, 0, 1, 2)$	$Q(2, -1, 1, 0, 2, 2)$	$Q(2, -1, 1, 1, 1, 1)$	$Q(2, -1, 1, 1, 1, 2)$	0.0017 0.0008 0.0021 0.0018
$Q(2, -1, 1, 2, 1, 2)$	$Q(2, -1, 1, 2, 2, 2)$	$Q(2, -1, 2, 0, 0, 0)$	$Q(2, -1, 2, 0, 0, 2)$	0.0009 0.0008 0.0021 0.0009
$Q(2, -1, 2, 0, 1, 0)$	$Q(2, -1, 2, 0, 1, 1)$	$Q(2, -1, 2, 0, 1, 2)$	$Q(2, -1, 2, 0, 2, 2)$	0.0022 0.0010 0.0021 0.0018
$Q(2, -1, 2, 1, 1, 1)$	$Q(2, -1, 2, 1, 1, 2)$	$Q(2, -1, 2, 2, 1, 2)$	$Q(2, -1, 2, 2, 2, 2)$	0.0020 0.0020 0.0018 0.0015
$Q(2, 0, -1, 0, 0, 0)$	$Q(2, 0, -1, 0, 0, 2)$	$Q(2, 0, -1, 0, 1, 0)$	$Q(2, 0, -1, 0, 1, 1)$	0.0013 0.0009 0.0021 0.0019
$Q(2, 0, -1, 0, 1, 2)$	$Q(2, 0, -1, 0, 2, 2)$	$Q(2, 0, -1, 1, 1, 1)$	$Q(2, 0, -1, 1, 1, 2)$	0.0013 0.0017 0.0022 0.0020

Continued on next page

Table S.3: Distribution of Q (continued).

Events ($a_0, a_1, a_2, b_0, b_1, b_2$)		Probabilities assigned					
$Q(2, 0, -1, 2, 1, 2)$	$Q(2, 0, -1, 2, 2, 2)$	$Q(2, 0, 0, 0, 0, 0)$	$Q(2, 0, 0, 0, 0, 2)$	0.0012	0.0015	0.0019	0.0012
$Q(2, 0, 0, 1, 0)$	$Q(2, 0, 0, 1, 1)$	$Q(2, 0, 0, 0, 1, 2)$	$Q(2, 0, 0, 0, 2, 2)$	0.0021	0.0008	0.0022	0.0020
$Q(2, 0, 1, 1, 1)$	$Q(2, 0, 1, 1, 2)$	$Q(2, 0, 0, 2, 1, 2)$	$Q(2, 0, 0, 2, 2, 2)$	0.0014	0.0020	0.0008	0.0022
$Q(2, 0, 1, 0, 0)$	$Q(2, 0, 1, 0, 2)$	$Q(2, 0, 1, 0, 1, 0)$	$Q(2, 0, 1, 0, 1, 1)$	0.0019	0.0015	0.0014	0.0021
$Q(2, 0, 1, 0, 1, 2)$	$Q(2, 0, 1, 0, 2, 2)$	$Q(2, 0, 1, 1, 1, 1)$	$Q(2, 0, 1, 1, 1, 2)$	0.0021	0.0011	0.0017	0.0013
$Q(2, 0, 1, 2, 1, 2)$	$Q(2, 0, 1, 2, 2, 2)$	$Q(2, 0, 2, 0, 0, 0)$	$Q(2, 0, 2, 0, 0, 2)$	0.0022	0.0007	0.0019	0.0016
$Q(2, 0, 2, 0, 1, 0)$	$Q(2, 0, 2, 0, 1, 1)$	$Q(2, 0, 2, 0, 1, 2)$	$Q(2, 0, 2, 0, 2, 2)$	0.0015	0.0016	0.0020	0.0015
$Q(2, 0, 2, 1, 1, 1)$	$Q(2, 0, 2, 1, 1, 2)$	$Q(2, 0, 2, 2, 1, 2)$	$Q(2, 0, 2, 2, 2, 2)$	0.0017	0.0015	0.0019	0.0012
$Q(2, 1, -1, 0, 0, 0)$	$Q(2, 1, -1, 0, 0, 2)$	$Q(2, 1, -1, 0, 1, 0)$	$Q(2, 1, -1, 0, 1, 1)$	0.0014	0.0016	0.0007	0.0017
$Q(2, 1, -1, 0, 1, 2)$	$Q(2, 1, -1, 0, 2, 2)$	$Q(2, 1, -1, 1, 1, 1)$	$Q(2, 1, -1, 1, 1, 2)$	0.0017	0.0021	0.0017	0.0021
$Q(2, 1, -1, 2, 1, 2)$	$Q(2, 1, -1, 2, 2, 2)$	$Q(2, 1, 0, 0, 0, 0)$	$Q(2, 1, 0, 0, 0, 2)$	0.0014	0.0010	0.0013	0.0020
$Q(2, 1, 0, 0, 1, 0)$	$Q(2, 1, 0, 0, 1, 1)$	$Q(2, 1, 0, 0, 1, 2)$	$Q(2, 1, 0, 0, 2, 2)$	0.0019	0.0016	0.0009	0.0016
$Q(2, 1, 0, 1, 1, 1)$	$Q(2, 1, 0, 1, 1, 2)$	$Q(2, 1, 0, 2, 1, 2)$	$Q(2, 1, 0, 2, 2, 2)$	0.0022	0.0022	0.0020	0.0017
$Q(2, 1, 0, 0, 0)$	$Q(2, 1, 0, 0, 2)$	$Q(2, 1, 1, 0, 1, 0)$	$Q(2, 1, 1, 0, 1, 1)$	0.0018	0.0019	0.0019	0.0017
$Q(2, 1, 1, 0, 1, 2)$	$Q(2, 1, 1, 0, 2, 2)$	$Q(2, 1, 1, 1, 1, 1)$	$Q(2, 1, 1, 1, 1, 2)$	0.0017	0.0021	0.0011	0.0021
$Q(2, 1, 2, 1, 2)$	$Q(2, 1, 2, 2, 2)$	$Q(2, 1, 2, 0, 0, 0)$	$Q(2, 1, 2, 0, 0, 2)$	0.0021	0.0011	0.0006	0.0015
$Q(2, 1, 2, 0, 1, 0)$	$Q(2, 1, 2, 0, 1, 1)$	$Q(2, 1, 2, 0, 1, 2)$	$Q(2, 1, 2, 0, 2, 2)$	0.0017	0.0014	0.0020	0.0017
$Q(2, 1, 2, 1, 1, 1)$	$Q(2, 1, 2, 1, 1, 2)$	$Q(2, 1, 2, 2, 1, 2)$	$Q(2, 1, 2, 2, 2, 2)$	0.0016	0.0014	0.0021	0.0011
$Q(2, 2, -1, 0, 0, 0)$	$Q(2, 2, -1, 0, 0, 2)$	$Q(2, 2, -1, 0, 1, 0)$	$Q(2, 2, -1, 0, 1, 1)$	0.0016	0.0006	0.0011	0.0016
$Q(2, 2, -1, 0, 1, 2)$	$Q(2, 2, -1, 0, 2, 2)$	$Q(2, 2, -1, 1, 1, 1)$	$Q(2, 2, -1, 1, 1, 2)$	0.0019	0.0019	0.0019	0.0020
$Q(2, 2, -1, 2, 1, 2)$	$Q(2, 2, -1, 2, 2, 2)$	$Q(2, 2, 0, 0, 0, 0)$	$Q(2, 2, 0, 0, 0, 2)$	0.0019	0.0022	0.0011	0.0009
$Q(2, 2, 0, 0, 1, 0)$	$Q(2, 2, 0, 0, 1, 1)$	$Q(2, 2, 0, 0, 1, 2)$	$Q(2, 2, 0, 0, 2, 2)$	0.0011	0.0016	0.0020	0.0008
$Q(2, 2, 0, 1, 1, 1)$	$Q(2, 2, 0, 1, 1, 2)$	$Q(2, 2, 0, 2, 1, 2)$	$Q(2, 2, 0, 2, 2, 2)$	0.0016	0.0010	0.0011	0.0011
$Q(2, 2, 1, 0, 0, 0)$	$Q(2, 2, 1, 0, 0, 2)$	$Q(2, 2, 1, 0, 1, 0)$	$Q(2, 2, 1, 0, 1, 1)$	0.0019	0.0019	0.0020	0.0006
$Q(2, 2, 1, 0, 1, 2)$	$Q(2, 2, 1, 0, 2, 2)$	$Q(2, 2, 1, 1, 1, 1)$	$Q(2, 2, 1, 1, 1, 2)$	0.0019	0.0010	0.0023	0.0019
$Q(2, 2, 1, 2, 1, 2)$	$Q(2, 2, 1, 2, 2, 2)$	$Q(2, 2, 2, 0, 0, 0)$	$Q(2, 2, 2, 0, 0, 2)$	0.0016	0.0015	0.0021	0.0016
$Q(2, 2, 2, 0, 1, 0)$	$Q(2, 2, 2, 0, 1, 1)$	$Q(2, 2, 2, 0, 1, 2)$	$Q(2, 2, 2, 0, 2, 2)$	0.0012	0.0010	0.0021	0.0017
$Q(2, 2, 2, 1, 1, 1)$	$Q(2, 2, 2, 1, 1, 2)$	$Q(2, 2, 2, 2, 1, 2)$	$Q(2, 2, 2, 2, 2, 2)$	0.0019	0.0023	0.0018	0.0010

Events (y_0, y_1, d_0, d_1)				Probabilities assigned			
$Q(0, 0, 0, 0)$	$Q(0, 0, 0, 1)$	$Q(0, 0, 1, 1)$	$Q(0, 1, 0, 0)$	0.0094	0.0001	0.0001	0.0001
$Q(0, 1, 0, 1)$	$Q(0, 1, 1, 1)$	$Q(0, 2, 0, 0)$	$Q(0, 2, 0, 1)$	0.0001	0.0001	0.0001	0.0001
$Q(0, 2, 1, 1)$	$Q(1, 0, 0, 0)$	$Q(1, 0, 0, 1)$	$Q(1, 0, 1, 1)$	0.0001	0.0001	0.0161	0.0001
$Q(1, 1, 0, 0)$	$Q(1, 1, 0, 1)$	$Q(1, 1, 1, 1)$	$Q(1, 2, 0, 0)$	0.0049	0.0024	0.0001	0.0001
$Q(1, 2, 0, 1)$	$Q(1, 2, 1, 1)$	$Q(2, 0, 0, 0)$	$Q(2, 0, 0, 1)$	0.0037	0.0200	0.0001	0.0001
$Q(2, 0, 1, 1)$	$Q(2, 1, 0, 0)$	$Q(2, 1, 0, 1)$	$Q(2, 1, 1, 1)$	0.0100	0.0116	0.0129	0.0001
$Q(2, 2, 0, 0)$	$Q(2, 2, 0, 1)$	$Q(2, 2, 1, 1)$		0.0001	0.9072	0.0002	

Table S.4: Distribution of Q_{3Y} .

Events (y_0, y_1, d_0, d_1)					Probabilities assigned						
$Q(0, 0, 0, 0)$	$Q(0, 0, 0, 1)$	$Q(0, 0, 1, 1)$	$Q(0, 1, 0, 0)$	$Q(0, 1, 0, 1)$	$Q(0, 1, 1, 1)$	0.0001	0.0142	0.0026	0.0001	0.0055	0.0164
$Q(0, 2, 0, 0)$	$Q(0, 2, 0, 1)$	$Q(0, 2, 1, 1)$	$Q(0, 3, 0, 0)$	$Q(0, 3, 0, 1)$	$Q(0, 3, 1, 1)$	0.0001	0.0595	0.0001	0.0001	0.0001	0.0004
$Q(1, 0, 0, 0)$	$Q(1, 0, 0, 1)$	$Q(1, 0, 1, 1)$	$Q(1, 1, 0, 0)$	$Q(1, 1, 0, 1)$	$Q(1, 1, 1, 1)$	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
$Q(1, 2, 0, 0)$	$Q(1, 2, 0, 1)$	$Q(1, 2, 1, 1)$	$Q(1, 3, 0, 0)$	$Q(1, 3, 0, 1)$	$Q(1, 3, 1, 1)$	0.0001	0.0056	0.0004	0.0069	0.0001	0.0043
$Q(2, 0, 0, 0)$	$Q(2, 0, 0, 1)$	$Q(2, 0, 1, 1)$	$Q(2, 1, 0, 0)$	$Q(2, 1, 0, 1)$	$Q(2, 1, 1, 1)$	0.0001	0.0025	0.0001	0.0001	0.0001	0.0001
$Q(2, 2, 0, 0)$	$Q(2, 2, 0, 1)$	$Q(2, 2, 1, 1)$	$Q(2, 3, 0, 0)$	$Q(2, 3, 0, 1)$	$Q(2, 3, 1, 1)$	0.0001	0.0001	0.0064	0.0161	0.0001	0.0001
$Q(3, 0, 0, 0)$	$Q(3, 0, 0, 1)$	$Q(3, 0, 1, 1)$	$Q(3, 1, 0, 0)$	$Q(3, 1, 0, 1)$	$Q(3, 1, 1, 1)$	0.0010	0.0530	0.0001	0.0001	0.0001	0.0001
$Q(3, 2, 0, 0)$	$Q(3, 2, 0, 1)$	$Q(3, 2, 1, 1)$	$Q(3, 3, 0, 0)$	$Q(3, 3, 0, 1)$	$Q(3, 3, 1, 1)$	0.0048	0.0001	0.0002	0.0001	0.7970	0.0003

Table S.5: Distribution of Q_{4Y} .